

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 24 — Electrochemistry

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Electrolysis · Faraday's Laws · Standard Electrode Potentials E° · Nernst Equation · Fuel Cells
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED**24.1 Quantitative Electrolysis & Faraday's Laws**— Electrolysis of molten and aqueous electrolytes; Faraday constant $F = Le$; quantitative relationship**24.2 Electrode Potentials & Electrochemical Cells**— Standard electrode potentials E° and standard cell potentials E°_{cell} ; Standard Hydrogen Electrode**24.3 Nernst Equation & Concentration Cells**— Nernst equation $E = E^\circ + (0.059/z) \log([oxidised]/[reduced])$; effect of non-standard concentration**24.4 Fuel Cells & Industrial Applications**

— Hydrogen-oxygen fuel cells in acidic and alkaline electrolytes; electrode half-equations; direct methanol

High-Frequency Core Repeats (Q41–Q50)

— The 10 most frequently tested Cambridge Paper 4 questions on Electrochemistry from the past 10

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Quantitative Electrolysis & Avogadro Constant Determination — 9701/42/M/J/23/Q2 [6 Marks]

HARD · Syllabus 24.1

A constant current of 0.850 A is passed through aqueous copper(II) sulfate for exactly 40.0 minutes using the apparatus shown in Fig. 1.1.

During electrolysis, copper metal is deposited on the cathode.

Data: $A_r(\text{Cu}) = 63.5$; Elementary charge $e = 1.602 \times 10^{-19} \text{ C}$; $F = 96500 \text{ C mol}^{-1}$.

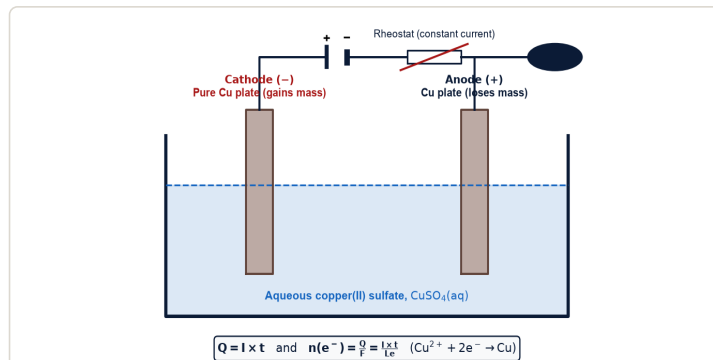


Fig. 1.1: Quantitative electrolysis circuit for determining mass of copper deposited at the cathode.

- (a) State the relationship connecting the Faraday constant (F), the Avogadro constant (L), and the elementary charge (e), and define the Faraday constant. [2]

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- (b) Calculate the mass of copper deposited at the cathode. [2]

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- (c) In an accurate experiment, 0.671 g of copper was deposited by 2040 C of charge. Use these data to calculate an experimental value for the Avogadro constant, L . [2]

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2 Standard Electrochemical Cell & Salt Bridge Dynamics — 9701/41/M/J/23/Q3 [6 Marks]

HARD · Syllabus 24.2

The standard electrochemical cell in Fig. 2.1 is set up under standard conditions of 298 K and 1.00 mol dm^{-3} .

Standard electrode potentials:

$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s})$, $E^\circ = -0.76 \text{ V}$

$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$, $E^\circ = +0.34 \text{ V}$

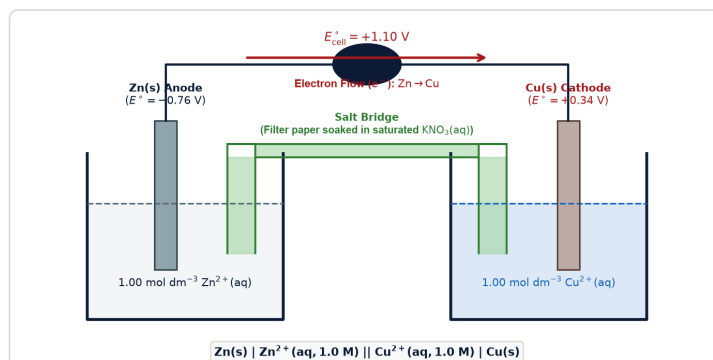


Fig. 2.1: Standard galvanic cell consisting of zinc and copper half-cells joined by a salt bridge.

- (a) Calculate the standard cell potential, E°_{cell} , and write the overall cell equation including state symbols. [2]
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- (b) Identify the negative electrode (anode) and state the direction of electron flow in the external circuit. [2]
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- (c) Explain the essential role of the salt bridge, and state which ions migrate into the copper half-cell as the cell discharges. [2]
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3 Nernst Equation Applied to Non-Standard Copper Half-Cell — 9701/42/O/N/23/Q2 [6 Marks]

HARD · Syllabus 24.3

The electrode potential of a metal half-cell depends on the concentration of aqueous metal ions according to the Nernst equation:

$$E = E^\circ + (0.059 / z) \log[\text{oxidised form}]$$

For copper: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$, $E^\circ = +0.34 \text{ V}$.

- (a) State the value of z in the Nernst equation for the copper half-cell, and calculate the electrode potential E when $[\text{Cu}^{2+}] = 0.020 \text{ mol dm}^{-3}$ at 298 K. [3]
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- (b) This copper half-cell is connected to a standard silver half-cell: $\text{Ag}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Ag}(\text{s})$, $E^\circ = +0.80 \text{ V}$. Calculate the cell potential, E_{cell} . [3]
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4 Standard Hydrogen Electrode (SHE) & Chromium Potential — 9701/41/O/N/23/Q2 [6 Marks]

HARD · Syllabus 24.2

The Standard Hydrogen Electrode (SHE) is the universal reference electrode shown in Fig. 4.1.

By definition, its standard electrode potential is 0.00 V.

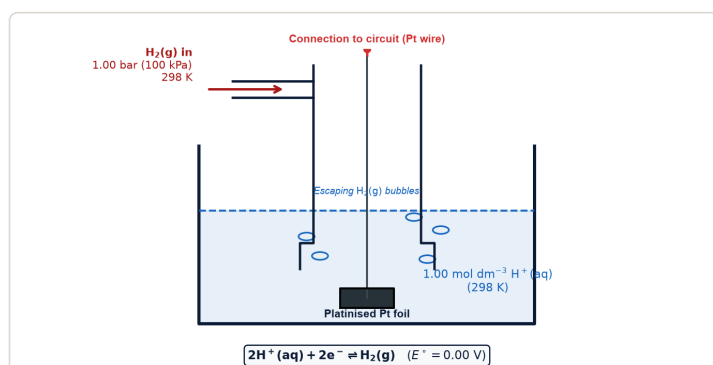


Fig. 4.1: The Standard Hydrogen Electrode (SHE) apparatus operating under standard conditions.

- (a) State the three standard conditions under which the SHE operates. [3]
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- (b) Explain the role of the platinised platinum foil in the SHE. [1]
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- (c) When the SHE is connected to a standard $\text{Cr}^{3+}(\text{aq})/\text{Cr}(\text{s})$ half-cell, the voltmeter reading is 0.74 V and electrons flow from the chromium electrode to the SHE. Deduce the value of E° for Cr^{3+}/Cr and write the half-equation occurring at the chromium electrode. [2]
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5 Alkaline Hydrogen–Oxygen Fuel Cell Kinetics & Thermodynamics — 9701/43/M/J/23/Q2 [6 Marks]

HARD · Syllabus 24.4

A schematic diagram of an alkaline hydrogen–oxygen fuel cell is shown in Fig. 5.1.

The electrolyte is hot aqueous potassium hydroxide, $\text{KOH}(\text{aq})$.

Data: E° values in alkaline solution:

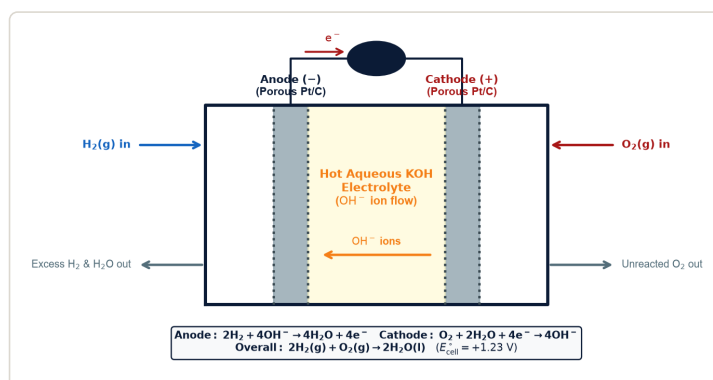
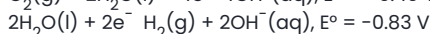
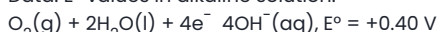


Fig. 5.1: Structure of an alkaline hydrogen–oxygen fuel cell using porous platinum–catalysed electrodes.

- (a) Write the half-equation occurring at the anode and the half-equation occurring at the cathode. [2]
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- (b) Calculate the standard cell potential, E°_{cell} , and write the overall cell reaction. [2]
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- (c) State two major advantages of a hydrogen–oxygen fuel cell over an internal combustion engine burning petrol. [2]
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6 Silver Electroplating & Faraday Constant Precision — 9701/42/M/J/22/Q3 [6 Marks]

HARD · Syllabus 24.1

A metal trophy is electroplated with silver using an aqueous solution of silver nitrate, $\text{AgNO}_3(\text{aq})$.

A current of 1.25 A is passed for 1 hour 15 minutes.

$A_r(\text{Ag}) = 107.9$; $F = 96500 \text{ C mol}^{-1}$.

- (a)** Write the half-equation for the reduction of silver ions at the cathode. **[1]**

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- (b)** Calculate the total quantity of charge passed and the theoretical mass of silver deposited. **[3]**

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- (c)** If the actual mass of silver deposited was 5.98 g, calculate the percentage current efficiency of the plating process. **[2]**

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7 Cobalt Complex Ligand Exchange Effect on Electrode Potential — 9701/41/M/J/22/Q3 [6 Marks]

HARD · Syllabus 24.2

Standard electrode potentials for cobalt(III)/cobalt(II) systems:

$[\text{Co}(\text{H}_2\text{O})_6]^{3+} + \text{e}^- \rightleftharpoons [\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $E^\circ = +1.82 \text{ V}$

$[\text{Co}(\text{NH}_3)_6]^{3+} + \text{e}^- \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{2+}$, $E^\circ = +0.10 \text{ V}$

$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}(\text{l})$, $E^\circ = +1.23 \text{ V}$

- (a)** Explain why the electrode potential becomes significantly less positive when water ligands are replaced by ammonia ligands. **[2]**

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- (b)** Deduce whether aqueous $\text{Co}^{2+}(\text{aq})$ can be oxidized to $\text{Co}^{3+}(\text{aq})$ by atmospheric oxygen under standard conditions. **[2]**

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- (c)** Deduce whether $[\text{Co}(\text{NH}_3)_6]^{2+}$ can be oxidized to $[\text{Co}(\text{NH}_3)_6]^{3+}$ by atmospheric oxygen, and calculate E°_{cell} for this reaction. **[2]**

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8 Dichromate(VI) / Iodide Redox Cell & pH Sensitivity — 9701/42/O/N/22/Q2 [6 Marks]

HARD · Syllabus 24.2

Electrode potentials:

$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$, $E^\circ = +1.33 \text{ V}$

$\text{I}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{I}^-(\text{aq})$, $E^\circ = +0.54 \text{ V}$

- (a) Construct the overall balanced ionic equation for the reaction that occurs when these two half-cells are connected. [2]
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- (b) Calculate the standard cell potential, E°_{cell} , and state what change in color would be observed in the chromium half-cell. [2]
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- (c) Predict and explain the effect on the cell potential if the pH of the dichromate half-cell is increased by adding aqueous sodium hydroxide. [2]
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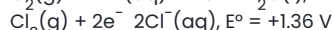
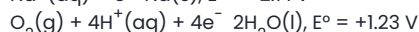
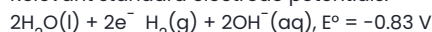
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Electrochemical Principles of the Chlor-Alkali Membrane Cell — 9701/41/O/N/22/Q3 [6 Marks]

HARD · Syllabus 24.1

In the industrial membrane cell, concentrated aqueous sodium chloride (brine) is electrolysed to produce chlorine gas, hydrogen gas, and aqueous sodium hydroxide.

Relevant standard electrode potentials:



- (a) Explain why hydrogen gas is discharged at the cathode instead of sodium metal. [2]
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- (b) Under standard conditions, oxygen has a less positive electrode potential (+1.23 V) than chlorine (+1.36 V). Explain why chlorine gas is evolved at the anode rather than oxygen in concentrated brine. [2]
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- (c) State the function of the cation-exchange membrane separating the anode and cathode compartments. [2]
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10

Copper Concentration Cell EMF & Nernst Derivation — 9701/43/O/N/22/Q2 [6 Marks]

HARD · Syllabus 24.3

A concentration cell is constructed using two copper electrodes immersed in copper(II) sulfate solutions of different concentrations, as shown in Fig. 10.1.

Beaker 1: $[\text{Cu}^{2+}] = 0.010 \text{ mol dm}^{-3}$

Beaker 2: $[\text{Cu}^{2+}] = 1.00 \text{ mol dm}^{-3}$

$E = E^\circ + (0.059 / z) \log[\text{Cu}^{2+}]$ at 298 K.

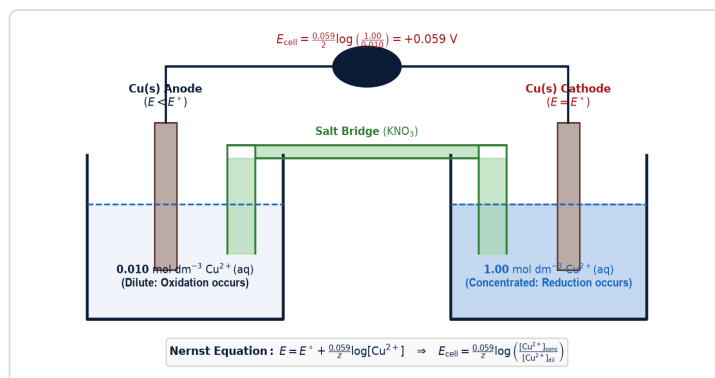


Fig. 10.1: Copper concentration cell driven by the concentration gradient between half-cells.

- (a) Calculate the electrode potential E of Beaker 1 and of Beaker 2, given $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$. [3]

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- (b) Calculate the overall cell potential, E_{cell} , and state which electrode acts as the cathode. [2]

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- (c) State what will happen to the cell potential when the cell is allowed to discharge over an extended period. [1]

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11

Chlorine Half-Cell Potential: Combined Concentration & Pressure Effects — 9701/42/M/J/21/Q3 [6 Marks] **HARD · Syllabus 24.3**

For the chlorine half-cell: $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq})$, $E^\circ = +1.36 \text{ V}$.

The general Nernst equation is:

$$E = E^\circ + (0.059 / z) \log([\text{oxidised}] / [\text{reduced}])$$

For gases, partial pressure in bar replaces concentration.

- (a) Write the specific Nernst expression for the chlorine electrode in terms of $p(\text{Cl}_2)$ and $[\text{Cl}^-]$. [2]

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- (b) Calculate the electrode potential E when $p(\text{Cl}_2) = 2.50 \text{ bar}$ and $[\text{Cl}^-] = 0.050 \text{ mol dm}^{-3}$ at 298 K . [3]

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- (c) State and explain whether chlorine becomes a stronger or weaker oxidizing agent under these non-standard conditions. [1]

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12 Industrial Electrefining of Blister Copper — 9701/41/M/J/21/Q3 [6 Marks]

HARD · Syllabus 24.1

Impure 'blister' copper contains small amounts of silver, gold, iron, and zinc.

During industrial electrolysis in aqueous CuSO_4 :

Standard potentials: $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$; $\text{Fe}^{2+}/\text{Fe} = -0.44 \text{ V}$; $\text{Cu}^{2+}/\text{Cu} = +0.34 \text{ V}$; $\text{Ag}^+/\text{Ag} = +0.80 \text{ V}$; $\text{Au}^{3+}/\text{Au} = +1.50 \text{ V}$.

(a) Explain what happens to the zinc and iron impurities during electrolysis. **[2]**

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(b) Explain what happens to the silver and gold impurities, and state their commercial significance. **[2]**

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(c) An electrolytic cell operates at 250 A for 24.0 hours. Calculate the mass of pure copper deposited on the cathode ($A_r(\text{Cu}) = 63.5$; $F = 96500 \text{ C mol}^{-1}$). **[2]**

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13 Proton Exchange Membrane (Acidic) Fuel Cell — 9701/43/M/J/21/Q2 [6 Marks]

HARD · Syllabus 24.4

A Proton Exchange Membrane Fuel Cell (PEMFC) operates in acidic medium.

Standard electrode potentials in acid solution:

$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$, $E^\circ = +1.23 \text{ V}$

$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$, $E^\circ = 0.00 \text{ V}$

(a) Write the half-equation at the anode and the half-equation at the cathode in this acidic fuel cell. **[2]**

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(b) Describe the path taken by electrons and the path taken by protons (H^+) as electrical work is produced. **[2]**

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(c) Calculate the standard Gibbs free energy change, G° , for the overall fuel cell reaction: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$. ($F = 96500 \text{ C mol}^{-1}$). **[2]**

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14 Copper(I) Disproportionation: Thermodynamic Calculation & Stability — 9701/42/O/N/21/Q3 [6 Marks]

HARD · Syllabus 24.2

Relevant standard electrode potentials:

$\text{Cu}^+(\text{aq}) + \text{e}^- \rightarrow \text{Cu}(\text{s})$, $E^\circ = +0.52 \text{ V}$

$\text{Cu}^{2+}(\text{aq}) + \text{e}^- \rightarrow \text{Cu}^+(\text{aq})$, $E^\circ = +0.15 \text{ V}$

(a) Write the ionic equation for the disproportionation of aqueous copper(I) ions. **[1]**

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- (b) Calculate the standard cell potential E°_{cell} for this disproportionation and deduce whether $\text{Cu}^+(\text{aq})$ is thermodynamically stable in water. [3]
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- (c) Explain why copper(I) iodide, $\text{CuI}(\text{s})$, is stable in contact with water and does not disproportionate ($K_{\text{sp}}(\text{CuI}) = 1.1 \times 10^{-12} \text{ mol}^2 \text{ dm}^{-6}$). [2]
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15

Electrolytic Evolution of Oxygen & Experimental Faraday Determination — 9701/41/O/N/21/Q2 [6 Marks]

HARD · Syllabus 24.1

Dilute sulfuric acid is electrolysed between inert platinum electrodes.

A current of 0.600 A passed for 45.0 minutes evolved 102 cm^3 of dry oxygen gas at the anode, measured at 298 K and $1.01 \times 10^5 \text{ Pa}$. Gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$.

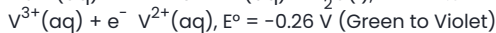
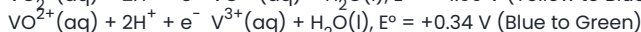
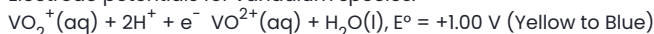
- (a) Write the ionic half-equation for the oxidation of water to form oxygen gas at the anode. [2]
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- (b) Calculate the number of moles of oxygen gas collected. [2]
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- (c) Calculate the total charge passed and determine an experimental value for the Faraday constant, F . [2]
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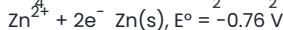
Vanadium Multistage Redox States & Selective Reduction — 9701/42/M/J/20/Q3 [6 Marks]

HARD · Syllabus 24.2

Electrode potentials for vanadium species:



Reducing agents:



- (a) Predict the final oxidation state and color of the vanadium solution when excess sulfur dioxide, $\text{SO}_2(\text{aq})$, is bubbled through acidified $\text{VO}_2^+(\text{aq})$. Justify with E° values. [3]
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- (b) Predict the final oxidation state and color when excess zinc metal is added to acidified $\text{VO}_2^+(\text{aq})$. Justify with E° values. [3]

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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Electrolytic Deposition of Tin from Tin(II) Sulfate — 9701/41/M/J/20/Q2 [4 Marks]

EASY · Syllabus 24.1

A solution of tin(II) sulfate, $\text{SnSO}_4(\text{aq})$, is electrolysed using a current of 1.60 A for 35.0 minutes.
 $A_r(\text{Sn}) = 118.7$; $F = 96500 \text{ C mol}^{-1}$.

- (a) Write the cathode half-equation and calculate the mass of tin deposited. [4]

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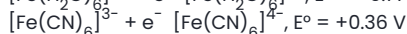
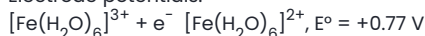
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18 Effect of Cyanide Complexation on Iron(III)/Iron(II) Potential — 9701/42/O/N/20/Q2 [4 Marks]

EASY · Syllabus 24.2

Electrode potentials:



- (a) Explain why the electrode potential is less positive for the hexacyanoferrate system than for the hydrated iron system. [4]

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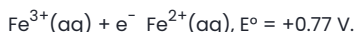
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19 Nernst Equation for Iron Half-Cell under Asymmetric Ion Ratios — 9701/41/O/N/20/Q3 [4 Marks]

EASY · Syllabus 24.3



Nernst equation: $E = E^\circ + 0.059 \log\left(\frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]}\right)$.

- (a) Calculate the electrode potential E when $[\text{Fe}^{3+}] = 0.50 \text{ mol dm}^{-3}$ and $[\text{Fe}^{2+}] = 0.0050 \text{ mol dm}^{-3}$ at 298 K. [4]

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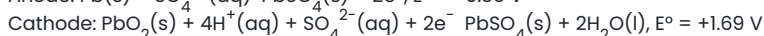
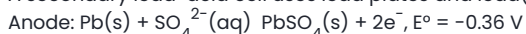
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20 Lead-Acid Storage Cell Discharge Electrochemistry — 9701/42/M/J/19/Q2 [4 Marks]

EASY · Syllabus 24.4

A secondary lead-acid cell uses lead plates and lead(IV) oxide in sulfuric acid.



- (a) Calculate the standard EMF of a single lead-acid cell and write the overall discharge equation. [4]

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21 Non-Standard Cell Potential of Zinc–Silver Cell — 9701/41/M/J/19/Q3 [4 Marks]

EASY · Syllabus 24.3

$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$, $E^\circ = -0.76 \text{ V}$; $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$, $E^\circ = +0.80 \text{ V}$.

A cell is assembled with $[\text{Zn}^{2+}] = 2.00 \text{ mol dm}^{-3}$ and $[\text{Ag}^+] = 0.010 \text{ mol dm}^{-3}$.

- (a) Calculate the non-standard potential of each half-cell and deduce the cell potential, E_{cell} . [4]

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22 Electrolysis of Lead(II) Bromide: Molten vs Aqueous — 9701/42/O/N/19/Q2 [4 Marks]

EASY · Syllabus 24.1

Lead(II) bromide, PbBr_2 , can be electrolysed in molten state or in saturated aqueous solution.

- (a) Contrast the products formed at the cathode and anode in molten PbBr_2 versus aqueous PbBr_2 , giving half-equations. [4]

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23 Direct Methanol Fuel Cell (DMFC) Electrochemistry — 9701/41/O/N/19/Q3 [4 Marks]

EASY · Syllabus 24.4

A direct methanol fuel cell uses liquid methanol, CH_3OH , as fuel.

Anode: $\text{CH}_3\text{OH(l)} + \text{H}_2\text{O(l)} \rightarrow \text{CO}_2\text{(g)} + 6\text{H}^+\text{(aq)} + 6\text{e}^-$

Cathode: $\text{O}_2\text{(g)} + 4\text{H}^+\text{(aq)} + 4\text{e}^- \rightarrow 2\text{H}_2\text{O(l)}$

- (a) Combine these equations to write the overall cell reaction and state two advantages of liquid methanol over hydrogen gas as fuel. [4]

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24 Function of Platinum Black Coating on Hydrogen Electrode — 9701/42/M/J/18/Q3 [4 Marks]

EASY · Syllabus 24.2

The platinum electrode in the SHE is coated with finely divided platinum black.

- (a) Explain two distinct scientific reasons why platinised platinum is used rather than a smooth platinum foil. [4]

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25 Avogadro Constant Calculation from Electrolysis of Silver — 9701/41/M/J/18/Q2 [4 Marks]

EASY · Syllabus 24.1

A current of 0.500 A was passed through $\text{AgNO}_3(\text{aq})$ for 38.6 minutes, depositing 1.295 g of silver.
 $e = 1.602 \times 10^{-19} \text{ C}$; $A_r(\text{Ag}) = 107.9$.

- (a) Calculate the value of the Avogadro constant, L , from these experimental results. [4]

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26 Nernst Shift Following Addition of Precipitating Halide — 9701/42/O/N/18/Q3 [4 Marks]

EASY · Syllabus 24.3

For $\text{Ag}^+(\text{aq}) + e^- \rightleftharpoons \text{Ag}(\text{s})$, $E^\circ = +0.80 \text{ V}$.

Excess $\text{NaCl}(\text{aq})$ is added to the silver half-cell, precipitating $\text{AgCl}(\text{s})$ ($K_{\text{sp}} = 1.8 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$).

- (a) Explain, using Le Chatelier's principle and the Nernst equation, why the electrode potential drops significantly. [4]

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27 Composition and Mechanism of the Salt Bridge — 9701/41/O/N/18/Q2 [4 Marks]

EASY · Syllabus 24.2

A standard galvanic cell utilizes a salt bridge saturated with potassium nitrate, $\text{KNO}_3(\text{aq})$.

- (a) Explain why KNO_3 is preferred over NaCl as salt bridge electrolyte, and explain how the bridge maintains electroneutrality. [4]

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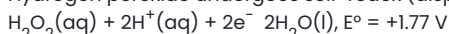
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28 Redox Disproportionation Feasibility of Hydrogen Peroxide — 9701/42/M/J/17/Q3 [4 Marks]

EASY · Syllabus 24.2

Hydrogen peroxide undergoes self-redox (disproportionation):



- (a) Calculate the standard cell potential E°_{cell} and explain why H_2O_2 is thermodynamically unstable with respect to disproportionation. [4]

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29 Electroplating Nickel from Nickel(II) Sulfate — 9701/41/M/J/17/Q2 [4 Marks]

EASY · Syllabus 24.1

An object is plated with a uniform layer of nickel from $\text{NiSO}_4(\text{aq})$ using a current of 2.50 A. $A_r(\text{Ni}) = 58.7$; Density of Ni = 8.90 g cm^{-3} ; $F = 96500 \text{ C mol}^{-1}$.

- (a) Calculate the time, in minutes, required to deposit 4.40 g of nickel. [4]

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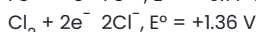
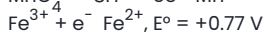
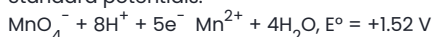
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30 Redox Titration Feasibility: Manganate(VII) vs Iron(II) — 9701/42/O/N/17/Q3 [4 Marks]

EASY · Syllabus 24.2

Standard potentials:



- (a) Calculate E°_{cell} for the titration of Fe^{2+} by MnO_4^- , and explain why hydrochloric acid cannot be used to acidify the titration. [4]

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31 Overpotential and Selective Discharge in Chlor-Alkali Cell — 9701/41/O/N/17/Q2 [4 Marks]

EASY · Syllabus 24.1

The discharge of ions at an electrode is governed by both thermodynamic electrode potentials and kinetic overpotentials.

- (a) Define 'overpotential' (overvoltage) and explain how it determines the discharge of chlorine rather than oxygen at a titanium anode. [4]

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32 Thermodynamics of the Daniell Cell & Maximum Electrical Work — 9701/42/M/J/16/Q3 [4 Marks]

EASY · Syllabus 24.2

The standard cell potential of the Daniell cell $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$ is +1.10 V at 298 K. $F = 96500 \text{ C mol}^{-1}$.

- (a) Calculate the standard Gibbs free energy change, G° , and state the maximum electrical work obtainable per mole of zinc consumed. [4]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Standard Definition of Electrode Potential — 9701/41/M/J/23/Q3(a) [2 Marks]

EASY · Syllabus 24.2

Electrode potential reflects electron-releasing or accepting tendency.

- (a) Define the standard electrode potential of a half-cell, E° . [2]

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34 Definition of the Faraday Constant and SI Units — 9701/42/M/J/23/Q2(a) [2 Marks]

EASY · Syllabus 24.1

Faraday constant links microscopic charge to molar quantities.

- (a) Define the Faraday constant, F , and state its SI units. [2]

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35 Operating Pressure and Concentration Standards in SHE — 9701/41/O/N/23/Q2(a) [2 Marks]

EASY · Syllabus 24.2

Standard conditions define reference potential reproducibility.

- (a) State the exact standard pressure of hydrogen gas and concentration of H^+ ions required in the Standard Hydrogen Electrode. [2]

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36 Statement of the Nernst Equation — 9701/42/O/N/23/Q2(a) [2 Marks]

EASY · Syllabus 24.3

Electrode potentials vary predictably with concentration.

- (a) State the mathematical form of the Nernst equation at 298 K, defining the term z . [2]

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37 Function of Inert Platinum in Redox Half-Cells — 9701/41/M/J/22/Q3(a) [2 Marks]

EASY · Syllabus 24.2

Half-cells involving two aqueous ions require an unreactive conductor.

- (a) Explain why a platinum electrode is used in an $\text{Fe}^{3+}(\text{aq})/\text{Fe}^{2+}(\text{aq})$ half-cell rather than an iron electrode. [2]

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38 Fundamental Relationship: Gibbs Energy and Cell EMF — 9701/42/M/J/22/Q3(a) [2 Marks]

EASY · Syllabus 24.2

Electrochemical cell potentials are directly linked to Gibbs free energy.

- (a) Write the thermodynamic equation relating standard Gibbs free energy change, G° , to standard cell potential, E°_{cell} . [2]

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39 Preferential Discharge Rule in Aqueous Electrolysis — 9701/41/O/N/22/Q3(a) [2 Marks]

EASY · Syllabus 24.1

Aqueous solutions contain competing cations and anions.

- (a) State the thermodynamic rule that predicts which cation is preferentially discharged at the cathode during electrolysis. [2]

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40 Thermodynamic Definition of Disproportionation — 9701/42/O/N/21/Q3(a) [2 Marks]

EASY · Syllabus 24.2

An intermediate oxidation state can undergo self-oxidation and self-reduction.

- (a) Define disproportionation in terms of oxidation numbers, and state the condition for its thermodynamic feasibility in terms of E°_{cell} . [2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 HF 1: Standard Hydrogen Electrode Comprehensive Analysis — 9701/42/M/J/23/Q3 [6 Marks]

HARD · Syllabus 24.2

HIGH-FREQUENCY CORE REPEAT #1 (Standard Hydrogen Electrode Mastery)

The Standard Hydrogen Electrode (SHE) shown in Fig. 41.1 is the primary international standard for electrode potential measurement.

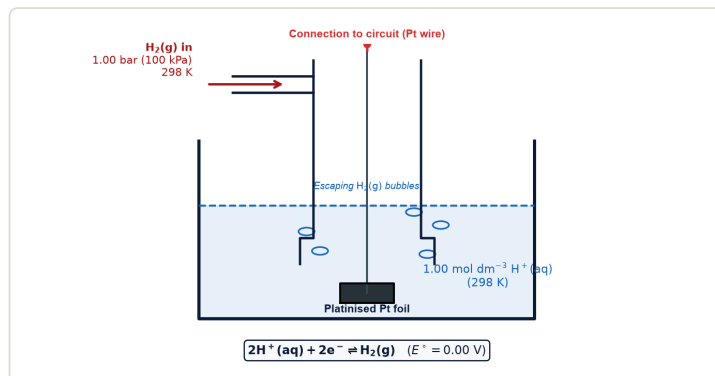


Fig. 41.1: Complete Standard Hydrogen Electrode (SHE) showing hydrogen gas inlet, platinised foil, and acid solution.

- (a) Label the components and define the exact operating conditions of temperature, pressure, and concentration in the SHE. [3]

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- (b) Describe how the SHE is used to measure the standard electrode potential of an unknown metal half-cell $M^{2+}(aq)/M(s)$. [3]

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42 HF 2: Daniell Cell Potentials, Salt Bridge Dynamics & Electron Flow — 9701/41/M/J/22/Q2 [6 Marks]

HARD · Syllabus 24.2

HIGH-FREQUENCY CORE REPEAT #2 (Electrochemical Cell & Salt Bridge Dynamics)

A zinc-copper electrochemical cell is set up as shown in Fig. 42.1.

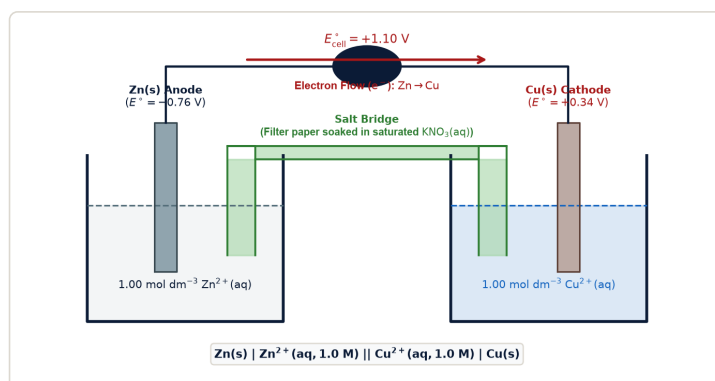
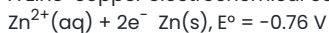


Fig. 42.1: Complete zinc-copper cell diagram showing voltmeter, salt bridge, and electron flow direction.

- (a) Calculate E°_{cell} , state the spontaneous cell reaction, and deduce which electrode loses mass. [3]

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- (b) Explain how the salt bridge maintains electrical continuity and explain the direction of migration of potassium and nitrate ions. [3]

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43 HF 3: Alkaline Hydrogen–Oxygen Fuel Cell Mechanics & Efficiency — 9701/42/O/N/22/Q3 [6 Marks]

HARD · Syllabus 24.4

HIGH-FREQUENCY CORE REPEAT #3 (Hydrogen–Oxygen Alkaline Fuel Cell)

The alkaline fuel cell in Fig. 43.1 uses concentrated KOH(aq) electrolyte and porous carbon electrodes containing platinum catalysts.

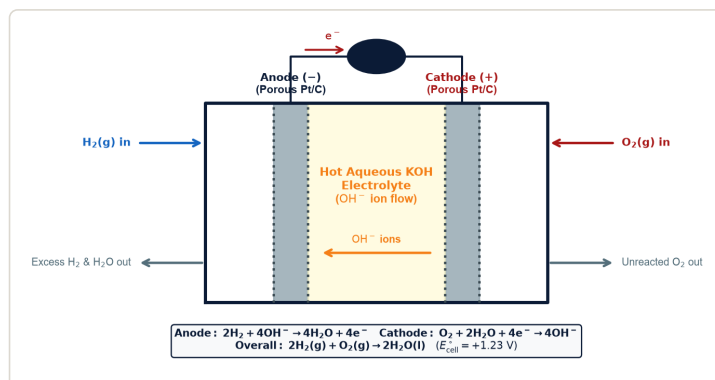


Fig. 43.1: Cross-section of an alkaline hydrogen–oxygen fuel cell showing gas inlets and electrolyte.

- (a) Write the balanced ionic half-equations occurring at the anode and at the cathode. [3]

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- (b) Calculate the standard cell potential E°_{cell} and explain why a fuel cell does not need to be recharged like a secondary battery. [3]

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44 HF 4: Quantitative Determination of Faraday and Avogadro Constants — 9701/41/O/N/21/Q3 [6 Marks]

HARD · Syllabus 24.1

HIGH-FREQUENCY CORE REPEAT #4 (Electrolytic Avogadro Determination)

In an electrolytic cell, an electric current of 0.400 A was passed through $\text{AgNO}_3(\text{aq})$ for 50.0 minutes.

The mass of silver deposited on the cathode was 1.341 g.

Data: $A_r(\text{Ag}) = 107.9$; Elementary charge on electron $e = 1.602 \times 10^{-19} \text{ C}$.

- (a) Calculate the total charge Q passed, and calculate the moles of electrons transferred. [2]

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- (b) Calculate an experimental value for the Faraday constant, F . [2]

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- (c) Using your calculated value of F and the value of e , determine the Avogadro constant, L .

[2]

45 **HF 5: Nernst Equation & Concentration Cell Thermodynamics — 9701/42/M/J/21/Q2 [4 Marks]**

EASY · Syllabus 24.3

HIGH-FREQUENCY CORE REPEAT #5 (Concentration Cell & Nernst Equation)

A copper concentration cell is illustrated in Fig. 45.1.

Left: $[\text{Cu}^{2+}] = 0.050 \text{ mol dm}^{-3}$; Right: $[\text{Cu}^{2+}] = 1.50 \text{ mol dm}^{-3}$.

$$E = E^\circ + (0.059 / 2) \log[\text{Cu}^{2+}].$$

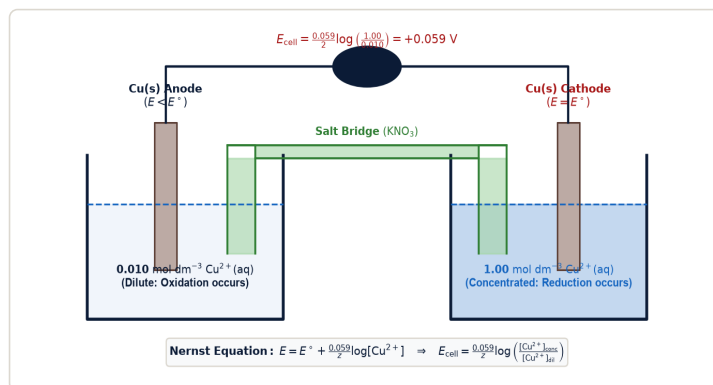


Fig. 45.1: Copper concentration cell driven by unequal Cu^{2+} ion concentrations.

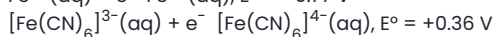
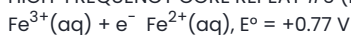
- (a) Calculate the potential difference E_{cell} of this concentration cell at 298 K.

[4]

46 **HF 6: Ligand Influence on Transition Metal Redox Stability — 9701/41/M/J/20/Q3 [4 Marks]**

EASY · Syllabus 24.2

HIGH-FREQUENCY CORE REPEAT #6 (Ligand Exchange & Redox Potentials)



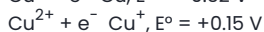
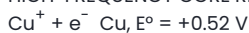
- (a) Explain how the nature of the ligand affects the relative stability of $\text{Fe}(\text{III})$ and $\text{Fe}(\text{II})$ and accounts for the difference in E° values.

[4]

47 **HF 7: Thermodynamic Feasibility of Copper(I) Disproportionation — 9701/42/O/N/20/Q3 [4 Marks]**

EASY · Syllabus 24.2

HIGH-FREQUENCY CORE REPEAT #7 (Disproportionation of Copper(I))



- (a) Use these data to prove that aqueous copper(I) sulfate disproportionates spontaneously into copper metal and copper(II) sulfate.

[4]

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48 HF 8: Selective Ion Discharge in Brine Electrolysis — 9701/41/O/N/19/Q2 [4 Marks]

EASY · Syllabus 24.1

HIGH-FREQUENCY CORE REPEAT #8 (Brine Electrolysis & Overpotentials)
Electrolysis of concentrated aqueous sodium chloride produces $\text{Cl}_2(\text{g})$ and $\text{H}_2(\text{g})$.

- (a) Explain why chlorine is liberated at the anode rather than oxygen, and write the anode and cathode half-equations.

[4]

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49 HF 9: Standard Definition of the Faraday Constant — 9701/41/M/J/23/Q2(a) [2 Marks]

EASY · Syllabus 24.1

HIGH-FREQUENCY CORE REPEAT #9 (Faraday Constant Definition)

- (a) Define the Faraday constant, F , and write its mathematical formula in terms of L and e .

[2]

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50 HF 10: Thermodynamic Criterion for Cell Feasibility — 9701/42/M/J/22/Q2(a) [2 Marks]

EASY · Syllabus 24.2

HIGH-FREQUENCY CORE REPEAT #10 (Thermodynamic Feasibility Criterion)

- (a) State the thermodynamic condition for a redox reaction to be feasible in terms of E°_{cell} , and state its relation to G° .

[2]

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Mark Scheme — Topic 24 — Electrochemistry

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	$F = L \times e$ [1]; The quantity of electric charge carried by one mole of electrons (or singly charged ions) [1].	2
1(b)	$Q = I \times t = 0.850 \times (40.0 \times 60) = 2040 \text{ C}$ [1]; Moles of Cu = $2040 / (2 \times 96500) = 0.01057 \text{ mol}$ & Arr; Mass = $0.01057 \times 63.5 = 0.671 \text{ g}$ [1].	2
1(c)	Moles of Cu = $0.671 / 63.5 = 0.010567 \text{ mol}$ [1]; Moles of $e^- = 2 \times 0.010567 = 0.02113 \text{ mol}$ & Arr; $L = Q / (n(e^-) \times e) = 2040 / (0.02113 \times 1.602 \times 10^{-19}) = 6.03 \times 10^{23} \text{ mol}^{-1}$ [1].	2
2(a)	$E^\circ_{\text{cell}} = E^\circ(\text{cathode}) - E^\circ(\text{anode}) = +0.34 - (-0.76) = +1.10 \text{ V}$ [1]; $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$ [1].	2
2(b)	Zn electrode is the negative electrode (anode) [1]; Electrons flow through external wire from Zn to Cu [1].	2
2(c)	Maintains electrical neutrality by allowing ion migration between half-cells / completes circuit without mixing solutions [1]; K^+ cations migrate into the copper half-cell to replace discharged Cu^{2+} ions [1].	2
3(a)	$z = 2$ [1]; $E = +0.34 + (0.059 / 2) \log(0.020)$ [1]; $\log(0.020) = -1.699$ & Arr; $E = +0.34 + 0.0295(-1.699) = +0.34 - 0.0501 = +0.29 \text{ V}$ [1].	3
3(b)	Ag half-cell remains standard so $E(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$ [1]; $E_{\text{cell}} = E(\text{cathode}) - E(\text{anode}) = E(\text{Ag}) - E(\text{Cu})$ [1]; $E_{\text{cell}} = +0.80 - (+0.29) = +0.51 \text{ V}$ [1].	3
4(a)	Temperature of 298 K (25°C) [1]; Hydrogen gas pressure of 1.00 bar (or 100 kPa / 1 atm) [1]; Hydrogen ion concentration $[\text{H}^+] = 1.00 \text{ mol dm}^{-3}$ [1].	3
4(b)	Provides an inert catalytic surface with high surface area for rapid electron transfer equilibrium between $\text{H}_2(\text{g})$ and $\text{H}^+(\text{aq})$ [1].	1
4(c)	Electrons flow from Cr & Arr; Cr is negative electrode & Arr; $E^\circ(\text{Cr}^{3+}/\text{Cr}) = -0.74 \text{ V}$ [1]; $\text{Cr(s)} \rightarrow \text{Cr}^{3+}(\text{aq}) + 3e^-$ [1].	2
5(a)	Anode: $2\text{H}_2(\text{g}) + 4\text{OH}^-(\text{aq}) \rightarrow 4\text{H}_2\text{O(l)} + 4e^-$ (or $\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2e^-$) [1]; Cathode: $\text{O}_2(\text{g}) + 2\text{H}_2\text{O(l)} + 4e^- \rightarrow 4\text{OH}^-(\text{aq})$ [1].	2
5(b)	$E^\circ_{\text{cell}} = +0.40 - (-0.83) = +1.23 \text{ V}$ [1]; $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O(l)}$ [1].	2
5(c)	Only product is water / zero greenhouse gas (CO_2) or NO_x emissions at point of use [1]; Higher thermodynamic energy conversion efficiency (not limited by Carnot cycle) [1].	2
6(a)	$\text{Ag}^+(\text{aq}) + e^- \rightarrow \text{Ag(s)}$ [1].	1
6(b)	$t = (1 \times 3600) + (15 \times 60) = 4500 \text{ s}$ [1]; $Q = I \times t = 1.25 \times 4500 = 5625 \text{ C}$ [1]; Mass of Ag = $(5625 / 96500) \times 107.9 = 0.05829 \times 107.9 = 6.29 \text{ g}$ [1].	3
6(c)	Current efficiency = (actual mass / theoretical mass) $\times 100$ [1]; Efficiency = $(5.98 / 6.29) \times 100 = 95.1\%$ [1].	2
7(a)	NH_3 is a stronger ligand / forms more stable complex with Co^{3+} than with Co^{2+} [1]; preferentially stabilizes the +3 oxidation state, shifting equilibrium to the left and making E° less positive [1].	2
7(b)	$E^\circ_{\text{cell}} = E^\circ(\text{O}_2) - E^\circ(\text{Co}^{3+}/\text{Co}^{2+}) = +1.23 - (+1.82) = -0.59 \text{ V}$ [1]; $E^\circ_{\text{cell}} < 0$ & Arr; reaction is not feasible under standard conditions [1].	2
7(c)	$E^\circ_{\text{cell}} = +1.23 - (+0.10) = +1.13 \text{ V}$ [1]; $E^\circ_{\text{cell}} > 0$ & Arr; feasible, $[\text{Co}(\text{NH}_3)_6]^{2+}$ is readily oxidized by air to $[\text{Co}(\text{NH}_3)_6]^{3+}$ [1].	2
8(a)	$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{I}^-(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{I}_2(\text{aq}) + 7\text{H}_2\text{O(l)}$ [2] (1 mark for correct species, 1 mark for balanced electrons).	2
8(b)	$E^\circ_{\text{cell}} = +1.33 - (+0.54) = +0.79 \text{ V}$ [1]; Color changes from orange ($\text{Cr}_2\text{O}_7^{2-}$) to green (Cr^{3+}) [1].	2
8(c)	Increasing pH decreases $[\text{H}^+]$ [1]; By Le Chatelier's principle / Nernst equation, equilibrium shifts left, making $E(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+})$ less positive & Arr; E_{cell} decreases [1].	2
9(a)	E° for reduction of H_2O (-0.83 V) is much more positive than E° for Na+ (-2.71 V) [1]; H_2O is vastly more easily reduced / requires less energy to gain electrons [1].	2
9(b)	High concentration of Cl^- ions in brine shifts the equilibrium / Nernst potential of $\text{Cl}_2/2\text{Cl}^-$ to a less positive value [1]; Oxidation of water to O_2 has a high activation energy (large kinetic overpotential) on the anode material [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
9(c)	Allows Na ⁺ ions to pass from anode to cathode compartment while preventing Cl ⁻ and OH ⁻ from crossing [1]; Prevents chlorine from reacting with OH ⁻ to form hypochlorite (or explosive H ₂ /Cl ₂ mixing) [1].	2
10(a)	Beaker 2 (standard): E = +0.34 V [1]; Beaker 1: E = +0.34 + (0.059/2) log(0.010) = +0.34 + 0.0295(-2) [1]; E(Beaker 1) = +0.34 - 0.059 = +0.281 V [1].	3
10(b)	E _{cell} = E(conc) - E(dil) = +0.340 - (+0.281) = +0.059 V (or via Nernst log(1.00/0.010)) [1]; The electrode in Beaker 2 (concentrated) is the cathode (reduction) [1].	2
10(c)	E _{cell} will decrease until it reaches 0.00 V when concentrations in both compartments become equal [1].	1
11(a)	z = 2 [1]; E = E° + (0.059 / 2) log(p(Cl ₂) / [Cl ⁻] ²) [1].	2
11(b)	p(Cl ₂) / [Cl ⁻] ² = 2.50 / (0.050) ² = 2.50 / 0.0025 = 1000 [1]; log(1000) = 3 [1]; E = +1.36 + (0.0295 × 3) = +1.36 + 0.0885 = +1.45 V [1].	3
11(c)	Stronger oxidizing agent because E is more positive (+1.45 V > +1.36 V) [1].	1
12(a)	Zn and Fe have more negative E° than Cu &Arr; easily oxidized at the anode into Zn ²⁺ (aq) and Fe ²⁺ (aq) ions [1]; They remain in solution because Cu ²⁺ is preferentially reduced at the cathode [1].	2
12(b)	Ag and Au have more positive E° than Cu &Arr; cannot be oxidized at the anode potential [1]; Fall to the bottom as 'anode sludge' / slime, recovered to offset refining costs [1].	2
12(c)	Q = 250 × (24 × 3600) = 2.16 × 10 ⁷ C [1]; Mass = (2.16 × 10 ⁷ / (2 × 96500)) × 63.5 = 111.9 × 63.5 = 7107 g (7.11 kg) [1].	2
13(a)	Anode: H ₂ (g) → 2H ⁺ (aq) + 2e ⁻ [1]; Cathode: O ₂ (g) + 4H ⁺ (aq) + 4e ⁻ → 2H ₂ O(l) [1].	2
13(b)	Electrons travel from anode through external circuit/load to cathode [1]; Protons permeate through the polymer electrolyte membrane directly to the cathode [1].	2
13(c)	n = 4 moles of electrons per mole of O ₂ (2 moles H ₂) [1]; G° = -nFE° _{cell} = -4 × 96500 × (+1.23) = -474780 J mol ⁻¹ = -475 kJ mol ⁻¹ [1].	2
14(a)	2Cu ⁺ (aq) → Cu ²⁺ (aq) + Cu(s) [1].	1
14(b)	Reduction: Cu ⁺ + e ⁻ → Cu (E° = +0.52 V) [1]; Oxidation: Cu ⁺ → Cu ²⁺ + e ⁻ (E° = +0.15 V) [1]; E° _{cell} = +0.52 - (+0.15) = +0.37 V &Arr; E° _{cell} > 0, so Cu ⁺ spontaneously disproportionates and is unstable in water [1].	3
14(c)	CuI is extremely insoluble, removing Cu ⁺ ions from solution so [Cu ⁺] is extremely small (~10 ⁻⁶ mol dm ⁻³) [1]; By Nernst equation, E(Cu ⁺ /Cu) drops below E(Cu ²⁺ /Cu ⁺), making disproportionation non-feasible / precipitation energetically stabilizes Cu(I) [1].	2
15(a)	2H ₂ O(l) → O ₂ (g) + 4H ⁺ (aq) + 4e ⁻ [2] (1 mark for correct species, 1 mark for balanced 4e ⁻).	2
15(b)	V = 102 × 10 ⁻⁶ m ³ [1]; n(O ₂) = pV / RT = (1.01 × 10 ⁵ × 102 × 10 ⁻⁶) / (8.31 × 298) = 10.302 / 2476.38 = 4.16 × 10 ⁻³ mol [1].	2
15(c)	Q = I × t = 0.600 × (45 × 60) = 1620 C [1]; 1 mol O ₂ requires 4 mol e ⁻ &Arr; Moles of e ⁻ = 4 × (4.16 × 10 ⁻³) = 0.01664 mol &Arr; F = Q / n(e ⁻) = 1620 / 0.01664 = 97350 C mol ⁻¹ [1].	2
16(a)	SO ₂ (E° = +0.17 V) can reduce VO ₂ ⁺ to VO ²⁺ (+1.00 - 0.17 = +0.83 V > 0) [1]; SO ₂ can reduce VO ₂ ⁺ to V ³⁺ (+0.34 - 0.17 = +0.17 V > 0) [1]; Cannot reduce V ³⁺ to V ²⁺ (-0.26 - 0.17 = -0.43 V < 0); Final state is V ³⁺ (green) [1].	3
16(b)	Zn (E° = -0.76 V) has a more negative potential than all three vanadium reduction steps [1]; Reduces VO ₂ ⁺ → VO ²⁺ → V ³⁺ → V ²⁺ sequentially because E° _{cell} > 0 for all steps [1]; Final state is V ²⁺ (violet) [1].	3
17(a)	Sn ²⁺ (aq) + 2e ⁻ → Sn(s) [1]; Q = 1.60 × (35 × 60) = 3360 C [1]; Moles of Sn = 3360 / (2 × 96500) = 0.01741 mol [1]; Mass of Sn = 0.01741 × 118.7 = 2.07 g [1].	4
18(a)	Cyanide CN ⁻ is a stronger ligand than H ₂ O [1]; Fe ³⁺ has higher charge density than Fe ²⁺ , attracting CN ⁻ more strongly [1]; [Fe(CN) ₆] ³⁻ is thermodynamically more stable relative to [Fe(CN) ₆] ⁴⁻ than Fe(H ₂ O) ₆ ³⁺ is to Fe(H ₂ O) ₆ ²⁺ [1]; Shifts the reduction equilibrium to the left, making E° less positive [1].	4
19(a)	Ratio [Fe ³⁺]/[Fe ²⁺] = 0.50 / 0.0050 = 100 [1]; log(100) = 2.0 [1]; E = +0.77 + (0.059 × 2.0) [1]; E = +0.77 + 0.118 = +0.89 V [1].	4
20(a)	E° _{cell} = +1.69 - (-0.36) = +2.05 V [2]; Pb(s) + PbO ₂ (s) + 2H ₂ SO ₄ (aq) → 2PbSO ₄ (s) + 2H ₂ O(l) [2].	4

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
21(a)	$E(\text{Zn}) = -0.76 + (0.059/2)\log(2.00) = -0.76 + 0.009 = -0.75 \text{ V}$ [1]; $E(\text{Ag}) = +0.80 + 0.059 \log(0.010) = +0.80 - 0.118 = +0.68 \text{ V}$ [1]; $E_{\text{cell}} = E(\text{Ag}) - E(\text{Zn})$ [1]; $E_{\text{cell}} = +0.68 - (-0.75) = +1.43 \text{ V}$ (vs $E^\circ = +1.56 \text{ V}$) [1].	4
22(a)	Molten cathode: $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}(\text{l})$ [1]; Molten anode: $2\text{Br}^- \rightarrow \text{Br}_2(\text{g}) + 2\text{e}^-$ [1]; Aqueous cathode: $\text{H}_2(\text{g})$ evolved ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$) or Pb deposited [1]; Aqueous anode: $\text{Br}_2(\text{aq})$ formed ($2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$) [1].	4
23(a)	$2\text{CH}_3\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$ [2]; Methanol is liquid at room temperature & much easier/safer to store and transport than compressed H_2 gas [1]; High volumetric energy density / existing liquid fuel distribution infrastructure can be adapted [1].	4
24(a)	Platinised platinum has an extremely porous, micro-rough surface providing a vastly increased surface area [2]; Acts as an effective heterogeneous catalyst for rapid dissociation of H_2 molecules into adsorbed H atoms [1]; Accelerates electron transfer equilibrium $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$, preventing polarization [1].	4
25(a)	$Q = 0.500 \times (38.6 \times 60) = 1158 \text{ C}$ [1]; Moles of Ag = $1.295 / 107.9 = 0.01200 \text{ mol}$ [1]; Moles of e^- = 0.01200 mol [1]; $L = Q / (n(\text{e}^-) \times e) = 1158 / (0.01200 \times 1.602 \times 10^{-19}) = 6.02 \times 10^{23} \text{ mol}^{-1}$ [1].	4
26(a)	Precipitation of AgCl removes Ag^+ ions from aqueous solution, dramatically reducing $[\text{Ag}^+]$ [1]; By Le Chatelier's principle, the equilibrium $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$ shifts to the left [1]; In Nernst equation $E = E^\circ + 0.059 \log[\text{Ag}^+]$, $\log[\text{Ag}^+]$ becomes large and negative [1]; Electrode potential E becomes significantly less positive (drops to approx +0.22 V) [1].	4
27(a)	K^+ and NO_3^- have virtually identical ionic mobilities, preventing liquid junction potentials [1]; Neither K^+ nor NO_3^- forms insoluble precipitates or redox reactions with typical cell electrolytes [1]; Anions (NO_3^-) migrate into the anode compartment to neutralize positive metal cations generated [1]; Cations (K^+) migrate into the cathode compartment to replace discharged cations [1].	4
28(a)	Reduction: $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$ (+1.77 V) [1]; Oxidation: $\text{H}_2\text{O}_2 \rightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ (+0.68 V) [1]; $E^\circ_{\text{cell}} = +1.77 - (+0.68) = +1.09 \text{ V}$ [1]; $E^\circ_{\text{cell}} > 0$ & thermodynamically feasible and spontaneous: $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ [1].	4
29(a)	Moles of Ni = $4.40 / 58.7 = 0.07496 \text{ mol}$ [1]; Moles of electrons = $2 \times 0.07496 = 0.1499 \text{ mol}$ [1]; $Q = n(\text{e}^-) \times F = 0.1499 \times 96500 = 14467 \text{ C}$ [1]; $t = Q / I = 14467 / 2.50 = 5787 \text{ s} = 96.4 \text{ minutes}$ [1].	4
30(a)	$E^\circ_{\text{cell}} = +1.52 - (+0.77) = +0.75 \text{ V}$ [1]; Reaction is completely feasible and quantitative [1]; MnO_4^- ($E^\circ = +1.52 \text{ V}$) has a more positive potential than Cl_2/Cl^- (+1.36 V) [1]; MnO_4^- would oxidize Cl^- ions from HCl into toxic Cl_2 gas, giving an erroneously high titre [1].	4
31(a)	Overpotential is the extra potential difference above the thermodynamic E° required to drive an electrode reaction at a measurable rate [2]; Oxidation of water to O_2 involves breaking multiple bonds and transfer of 4 electrons, conferring a very high activation energy and high overpotential [1]; Discharge of Cl^- involves simple electron loss with minimal overpotential on titanium, making chlorine kinetically favored [1].	4
32(a)	$n = 2$ [1]; $G^\circ = -nFE^\circ_{\text{cell}} = -2 \times 96500 \times 1.10$ [1]; $G^\circ = -212300 \text{ J mol}^{-1} = -212.3 \text{ kJ mol}^{-1}$ [1]; Maximum electrical work = $-G^\circ = 212.3 \text{ kJ mol}^{-1}$ [1].	4
33(a)	The electromotive force (EMF) of a cell composed of the half-cell connected to a standard hydrogen electrode [1] under standard conditions of 298 K, 100 kPa pressure, and 1.00 mol dm ⁻³ solution concentrations [1].	2
34(a)	The electric charge carried by one mole of electrons (or singly charged ions) [1]; Units: C mol ⁻¹ (coulombs per mole) [1].	2
35(a)	Pressure: 1.00 bar (or 100 kPa / 1 atm) [1]; Concentration: 1.00 mol dm ⁻³ $\text{H}^+(\text{aq})$ [1].	2
36(a)	$E = E^\circ + (0.059 / z) \log([\text{oxidised form}] / [\text{reduced form}])$ [1]; z is the number of electrons transferred in the half-equation [1].	2
37(a)	Platinum is chemically inert and does not react with the solution or participate in the redox reaction [1]; It provides a conducting surface for electron transfer between Fe^{3+} and Fe^{2+} (iron would dissolve reacting directly) [1].	2
38(a)	$G^\circ = -nFE^\circ_{\text{cell}}$ (or $G^\circ = -zFE^\circ_{\text{cell}}$) [1]; n = moles of electrons transferred, F = Faraday constant [1].	2
39(a)	The cation with the more positive (or less negative) standard electrode potential E° is preferentially reduced / discharged [2].	2
40(a)	A redox reaction in which atoms of the same element are simultaneously oxidized and reduced [1]; Feasible when $E^\circ(\text{reduction}) > E^\circ(\text{oxidation})$, meaning $E^\circ_{\text{cell}} > 0$ [1].	2
41(a)	Pt wire/foil coated with platinum black; 1.00 mol dm ⁻³ $\text{H}^+(\text{aq})$ (or 0.5 mol dm ⁻³ H_2SO_4) [1]; Pure H_2 gas bubbled at 1.00 bar (100 kPa) [1]; Maintained at 298 K (25 °C) [1].	3

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
41(b)	Connect SHE to M ²⁺ /M half-cell using a salt bridge [1]; Connect electrodes via a high-resistance digital voltmeter [1]; Voltmeter reading gives magnitude of E°; if M is negative electrode, E° is negative, if positive electrode, E° is positive [1].	3
42(a)	E°cell = +0.34 - (-0.76) = +1.10 V [1]; Zn(s) + Cu ²⁺ (aq) → Zn ²⁺ (aq) + Cu(s) [1]; Zn electrode loses mass (oxidized to Zn ²⁺) [1].	3
42(b)	Completes the electrical circuit by allowing movement of ions between solutions without direct mixing [1]; NO ₃ ⁻ anions move toward the Zn half-cell to balance newly formed Zn ²⁺ cations [1]; K ⁺ cations move toward the Cu half-cell to replace discharged Cu ²⁺ cations [1].	3
43(a)	Anode: 2H ₂ (g) + 4OH ⁻ (aq) → 4H ₂ O(l) + 4e ⁻ [2]; Cathode: O ₂ (g) + 2H ₂ O(l) + 4e ⁻ → 4OH ⁻ (aq) [1].	3
43(b)	E°cell = +0.40 - (-0.83) = +1.23 V [1]; Fuel (H ₂) and oxidant (O ₂) are supplied continuously from an external source [1]; Operates indefinitely as long as reactants are supplied, unlike batteries with finite stored chemical energy [1].	3
44(a)	Q = I × t = 0.400 × (50.0 × 60) = 1200 C [1]; Moles of Ag = 1.341 / 107.9 = 0.012428 mol [1]; Moles of e ⁻ = 0.012428 mol [1].	2
44(b)	F = Q / n(e ⁻) = 1200 / 0.012428 [1]; F = 96556 C mol ⁻¹ (accept 96560 C mol ⁻¹) [1].	2
44(c)	L = F / e = 96556 / (1.602 × 10 ⁻¹⁹) [1]; L = 6.027 × 10 ²³ mol ⁻¹ (accept 6.03 × 10 ²³) [1].	2
45(a)	Ecell = (0.059 / 2) log([Cu ²⁺]conc / [Cu ²⁺]dil) [1]; Ratio = 1.50 / 0.050 = 30.0 [1]; log(30.0) = 1.477 [1]; Ecell = 0.0295 × 1.477 = +0.0436 V (or +0.044 V) [1].	4
46(a)	CN ⁻ forms stronger coordinate bonds than H ₂ O [1]; Fe ³⁺ has smaller ionic radius and higher charge density than Fe ²⁺ [1]; Fe ³⁺ attracts CN ⁻ ligands more strongly, forming a thermodynamically more stable complex than Fe ²⁺ [1]; This preferentially stabilizes the oxidized form, shifting equilibrium to the left and reducing E° [1].	4
47(a)	Reduction half-equation: Cu ⁺ + e ⁻ → Cu (E° = +0.52 V) [1]; Oxidation half-equation: Cu → Cu ²⁺ + e ⁻ (E° = +0.15 V) [1]; Overall: 2Cu ⁺ (aq) → Cu ²⁺ (aq) + Cu(s) [1]; E°cell = +0.52 - (+0.15) = +0.37 V [1]; Since E°cell > 0, the reaction is thermodynamically spontaneous [1].	4
48(a)	Anode: 2Cl ⁻ (aq) → Cl ₂ (g) + 2e ⁻ [1]; Cathode: 2H ₂ O(l) + 2e ⁻ → H ₂ (g) + 2OH ⁻ (aq) (or 2H ⁺ + 2e ⁻ → H ₂) [1]; High concentration of Cl ⁻ in brine lowers its discharge potential [1]; Oxidation of water to O ₂ has a large kinetic activation energy / high overpotential on the anode surface [1].	4
49(a)	The electric charge carried by one mole of electrons [1]; F = L × e [1].	2
50(a)	E°cell must be positive (E°cell > 0) [1]; Corresponds to a negative Gibbs free energy change (G° < 0) via G° = -nFE°cell [1].	2